

Laypeople's and Researchers' Perspectives on Real-Life Risky Choices: A Comparative Analysis of Overlaps and Discrepancies

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To validate novel measurement instruments of people's decision making, the behavioral sciences often assess how frequently people make specific risky choices in real life. However, it remains unclear to what degree the risky choices tapped by such frequency measures reflect those choices people actually (have to) make. To address this issue, we compared 100 risky choices representing laypeople's perspective with 63 risky choices representing researchers' perspective. In Study 1, we leveraged natural language processing calibrated on human judgments to gauge the similarity between choices of both perspectives and found that they only had 18% of the choices in common. In Study 2, we further examined the implications of this mismatch in terms of the psychological mechanisms that the various choices may tap into by asking 825 participants to rate the perceived relevance of seven classes of psychological mechanisms to their decision making in these choices. Bayesian mixed effects models revealed credible differences between the two perspectives in five out of seven classes of mechanisms: For choices representing laypeople's perspective, choice attributes, time factors, experience and knowledge, and goals and motivation were on average perceived as more relevant, and social factors were perceived as less relevant, relative to the choices representing researchers' perspective. These findings suggest that decision-making paradigms calibrated on frequency measures from the researchers' perspective may have limited generalizability to the broad range of risky choices people face in their lives, underscoring the need to better understand the complexity of real-life decision making.

Keywords: decisions under risk and uncertainty, risky choice, data-driven inventory, psychological mechanisms, Bayesian analysis

Supplemental materials: <https://doi.org/10.1037/dec0000268.supp>

This article was published Online First January 12, 2026.
Tim Rakow served as action editor.

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The authors have no conflicts of interest to disclose. This work was supported by the Swiss National Science Foundation (Grant PCEFP1_194540 awarded to Renato Frey). Because this research was funded in whole, or in part, by Swiss National Science Foundation (Grant 194540), for the purpose of open access, the author has applied a CC BY public copyright license to any author accepted manuscript version arising from this submission.

The authors are grateful to Aita Ammann and Lennart Müller for their invaluable support in conducting the literature

search and classification of choices for the researchers' perspective as well as Sabou Rani Stocker for her help with additional analyses.

Olivia Fischer played a lead role in data curation, formal analysis, investigation, methodology, project administration, validation, visualization, and writing—original draft and an equal role in conceptualization, resources, and writing—review and editing. Aaron B. Lob played a supporting role in conceptualization, formal analysis, investigation, methodology, project administration, and writing—review and editing. Renato Frey played a lead role in funding acquisition and supervision, an equal role in conceptualization, resources, and writing—review and editing, and a supporting

continued

To develop and test theories on how people make decisions under risk, ambiguity, and uncertainty more generally,¹ the behavioral sciences have developed various measurement instruments to capture a person's risk preferences and related constructs (e.g., behavioral tasks and questionnaires of hypothetical scenarios; Lejuez et al., 2002; Weber et al., 2002). Ultimately, the aspiration is that these instruments allow researchers to better understand and predict real-life decision-making processes and outcomes, develop effective interventions, and inform decision-making policies on relevant societal issues (Clark & Lisowski, 2017; Frey, 2021; Markiewicz & Weber, 2013; Szrek et al., 2012).

Crucially, for such measurement instruments—and by extension, the theories and models tested with them—to generalize to consequential choices and behaviors in real life, they need to be calibrated appropriately. To this end, researchers have often used frequency measures of real-life risky choices. As the name suggests, frequency measures aim to capture (i.e., via self-reports) how often individuals engage or have previously engaged in specific risky behaviors outside laboratory settings (see Frey et al., 2017). For instance, participants may be asked “How often do you have a drink containing alcohol?” (Allen et al., 2001, p. 432) “How many cigarettes per day do you smoke?” (Heatherton et al., 1991, p. 1121) or how frequently they have “bet on horses, dogs, or other animals (in off-track betting, at the track, or with a bookie)” (Lesieur & Blume, 1987, p. 1187). Of note, the real-life risky choices tapped by these and similar frequency measures do not exclusively—or not even primarily—comprise decisions under risk in the narrower sense but also decisions under uncertainty more broadly (see Footnote 1). This suggests that at least regarding real-life contexts, researchers may use the term risky choice in a rather inclusive manner.

Applications of frequency measures of risky choices as tools to calibrate and validate specific measurement instruments are manifold. To mention just two examples, during the initial validation of the balloon analogue risk task (a behavioral measure with monetary incentives), it was tested whether the

task is associated with a number of real-life risky choices, including frequency of drinking, smoking, gambling, stealing, and sexual intercourse without the use of a condom (Lejuez et al., 2002). Similarly, during the development of the Domain-Specific Risk-Taking Scale (a propensity measure using hypothetical scenarios), frequency measures were used to assess whether specific subscales are associated with buying illegal drugs, binge drinking, and doing something dangerous on a dare (Weber et al., 2002). Consequently, the employment of specific tasks and questionnaires is often justified based on the argument that their external validity has been established using specific frequency measures during initial task development (e.g., Aklin et al., 2005; Fairley et al., 2019; Wilke et al., 2014).

Given the critical role frequency measures of real-life risky choices play in the process of calibrating and evaluating various measurement instruments of people's decision making under risk and uncertainty, the question arises: What are relevant risky choices that frequency measures ought to target?

Researchers' Perspective

One path to answering this question consists of adopting what we will refer to as researchers' perspective. Specifically, a systematic inspection of the extant frequency measures promises to reveal what behavioral researchers consider (or have previously considered) relevant real-life risky choices. Although frequency measures are naturally not the only means by which real-life

¹ The term *decision under risk* is often used to refer to choices in which the potential outcomes and their probabilities of occurrence are known (i.e., a situation of *a priori* probabilities or risk proper; Knight, 1921). Conversely, the term *uncertainty* (or ambiguity) is often used to refer to choices with unknown outcomes (or unknown outcome probabilities, e.g., Edwards, 1954; Knight, 1921). In the latter, outcomes and probabilities can only be inferred empirically (statistical probabilities) or by guessing (i.e., estimates; Knight, 1921). Whereas this distinction excludes risk from uncertainty, some researchers conceptualize risk as one of two types of uncertainty (i.e., aleatory uncertainty; Fox & Ülkümen, 2011).

role in formal analysis, investigation, methodology, and project administration.

 The data are available at <https://osf.io/f6e7q/>, <https://osf.io/qyfw8/>.

 The experimental materials are available at <https://osf.io/f6e7q/>, <https://osf.io/qyfw8/>.

 The preregistered design is available at <https://osf.io/8ubfm>, <https://osf.io/ms47q>.

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risky choices are studied (see, e.g., Gigerenzer, 2006; Klein, 2017; Siegrist & Cvetkovich, 2000; Slovic, 1987), an analysis of the risky choices tapped by the extant frequency measures may be particularly insightful given their importance during the development of many other measurement instruments assessing individual differences.

To obtain an insight into researchers' perspective on what constitutes relevant risky choices through this particular lens, we previously assembled an inventory of frequency measures commonly used in the behavioral sciences (<https://osf.io/nrp6u>). To do so, we first reviewed seminal literature on risk taking (e.g., Buelow & Suhr, 2009; Frey et al., 2017; Lejuez et al., 2002) and then conducted a literature search on Google Scholar with keywords such as "risk measure," "risk scale," and "risk frequency" as well as domain-specific keywords such as "traffic," "health," "occupational," "social," "alcohol," "drugs," "recreational," "eating," "driving," "sports," and "gambling." As a result, 620 candidate frequency measures were identified. Moreover, a subsequent online survey with researchers of the behavioral sciences resulted in the addition of 24 further frequency measures. Finally, we applied a systematic procedure of consolidating these choices (e.g., collapsing identical choices, only retaining choices that appeared in at least three frequency measures to prevent the inclusion of overly niche choices, excluding frequency measures that only capture precursors or consequences of a specific risky behavior).

The resulting inventory comprised 63 unique risky choices, many of which are health-related (e.g., smoke a cigarette), clinically relevant (e.g., inflict self-harm), or antisocial behaviors (e.g., destroy property). Interestingly, the composition of risky choices in this inventory may reflect that many frequency measures were originally developed in clinical or developmental contexts. This, in turn, suggests that researchers' perspective on what constitutes relevant real-life risky choices to be examined via frequency measures may be shaped by theoretical frameworks that have historically emphasized stereotypical or maladaptive choices (e.g., drug use, alcohol consumption, smoking; Duell & Steinberg, 2019; Verschoor et al., 2016). To illustrate, scales such as the Drug Abuse Screening Test (Skinner, 1982), the South Oaks Gambling Screen (Lesieur & Blume, 1987), and the Fagerström Test for Nicotine Dependence (Heatherton et al., 1991) are common frequency

measures of people's real-life choice behavior (e.g., Frey et al., 2017; Huang et al., 2024; Sirola et al., 2019). Clearly, the risky behaviors studied with these questionnaires remain highly relevant today and warrant continued research attention (e.g., by adapting an existing questionnaire for nicotine dependence to e-cigarette use; Piper et al., 2020). And yet, it is important to keep in mind that current frequency measures only reflect one of multiple perspectives on what constitutes relevant real-life risky choices.

Laypeople's Perspective

A second and alternative path to answering the question of what constitutes relevant risky choices in real life consists of examining laypeople's perspective on this matter. That is, rather than answering this question by focusing on the frequency measures developed in a top-down manner by researchers, it may also be highly informative to directly ask laypeople (i.e., bottom-up) about their encounters with risky choices. Whereas it is relatively straightforward to examine the real-life risky choices constituting researchers' perspective on this matter—because common frequency measures can readily be extracted from the literature—mapping the risky choices people actually encounter in real-life settings requires a more extensive, empirical approach.

We previously did so by conducting a series of large population surveys that directly asked laypeople about the occurrence of risky choices in their lives (Frey & Fischer, 2025). Specifically, 4,380 people of the Swiss general public reported either a risky choice that they had encountered personally in their own lives or a risky choice that they believed many fellow citizens have been confronted with (the latter being inspired by the idea of social sensing; Galesic et al., 2021). Moreover, to obtain as diverse a set of risky choices as possible, we also varied whether participants were asked to report a risky choice that they ultimately decided in favor of or against. As an example, participants were asked to report a risk that they themselves had taken, as follows: "Now please try to remember an important decision from your own life where you had the choice to either take a risk or avoid it—and you ultimately decided in favor of the risk." Importantly, in line with the rationale provided above, risk was left intentionally undefined for participants' own evaluation.

By means of a systematic classification approach, we identified unique and commonly reported risky

choices (e.g., excluding those that were reported fewer than three times; for methodological details, see Frey & Fischer, 2025). This data-driven approach ultimately resulted in an inventory of 100 real-life risky choices—thus providing a snapshot of the current ecology of risk and representing laypeople’s perspective on what constitutes relevant risky choices. Interestingly, this perspective contains many (but not exclusively) choices that are occupational (e.g., quit one’s job) and financial (e.g., invest money) in nature, thus at least at first glance implying some disconnect to the real-life choices typically focused on by researchers.

Overview of Aims and Research Questions

The researchers’ and laypeople’s perspectives on real-life risky choices each have their own merits—and there likely cannot be a definitive answer as to whether the current status quo of researchers’ or laypeople’s perspective ought to be the gold standard. However, and returning to the question of how best to calibrate the various measurement instruments of people’s risk preferences and related constructs, it is crucial to study whether and how researchers’ and laypeople’s perspectives coincide. Specifically, only under some conditions can one expect insights obtained from the extant measurement instruments—which were mostly calibrated on frequency measures from the researchers’ perspective—to generalize to the diverse risky choices laypeople are confronted with. Hence, in two studies we pursued the following two aims (both studies, including their design, materials, and analysis pipelines, were preregistered at <https://osf.io/f6e7q/>).

Aim 1: Comparing and Contextualizing Choices From Both Perspectives

First, if the two perspectives fully overlapped in the specific real-life risky choices they entail, one could arguably expect the highest possible generalizability from measurement in the lab to real life because the extant frequency measures used to calibrate other measurement instruments would essentially tap the same risky choices that people face in life. However, as briefly illustrated above, this may not be an entirely realistic scenario. It is therefore important to analyze to what extent the two perspectives involve nominally identical choices and how many of them are unique to each perspective.

To do so, in Study 1 we thus first quantified the semantic similarity between choices representing both perspectives—63 choices representing researchers’ perspective and 100 choices representing laypeople’s perspective, 163 choices in total—using a natural language processing approach. We then ran an empirical calibration study to identify a threshold of semantic similarity above which human raters consider any two choices identical. We thereby determined the overlapping and unique choices representing the two perspectives.

Moreover and using the same method of quantifying semantic similarity between choices, we studied how both perspectives relate to three external resources on hazards and risk-related behaviors: big life decisions (Camilleri, 2023), global hazards (World Economic Forum, 2024), and health-related hazards (Murray et al., 2020). This allows situating the two perspectives within a broader context: For instance, do the choices representing researchers’ perspective more closely relate to health hazards described in the Global Burden of Disease (GBD) repository (Murray et al., 2020)? Or do choices representing laypeople’s perspective more closely resemble important and big life decisions (Camilleri, 2023)?

Taken together, in Study 1 we addressed our first research question: To what extent do researchers’ and laypeople’s perspectives converge or diverge—and how do these choices fit into other conceptualizations of relevant choices and hazards? Addressing this question could (a) help in identifying potential blind spots that should be addressed in future developments of frequency measures and (b) contribute to a better understanding of how the two perspectives conceptualize relevant risky choices.

Aim 2: Analyzing Which Classes of Psychological Mechanisms the Risky Choices From Both Perspectives May Tap Into

Second, even if there is no perfect overlap between risky choices from the two perspectives, measurement instruments calibrated on the extant frequency measures may still generalize to real-life risky choices—provided that they tap into the same classes of psychological mechanisms. To illustrate, a specific real-life risky choice may be strongly influenced by the affective reaction a person anticipates to experience after learning about the choice’s outcomes (e.g., Loewenstein et al., 2001; Mellers et al., 1999). Thus, even if a risky choice of the researchers’ perspective is

nominally different from this real-life choice, insights from the former may still allow predicting behavior in the latter—to the extent that it is also strongly shaped by anticipated affective reactions.

With psychological mechanisms we refer to the causal relationships between specific psychological processes (e.g., perception, cognition, affect) and the mental states or behaviors they influence (e.g., Koch & Cratsley, 2020; Lob et al., 2025). A recent taxonomy (Lob et al., 2025) of such risk-related mechanisms identified 83 distinct mechanisms from a wide range of theories (e.g., Bernoulli, 1738; Kahneman & Tversky, 1979; Savage, 1954; Simon, 1955). These mechanisms clustered in seven major classes, including mechanisms related to the subjective evaluation and perception of choice attributes (e.g., transformations of values or probabilities; Bernoulli, 1738; Kahneman & Tversky, 1979; Savage, 1954; von Neumann & Morgenstern, 1944), cognitive resources and control (e.g., attention; Busemeyer & Townsend, 1993), social factors (e.g., accountability; Tetlock, 1992), affect (e.g., anticipated pleasure; Mellers et al., 1999), goals and motivation (e.g., need; Mishra et al., 2017), experience and knowledge (e.g., previous exposure; Loewenstein et al., 2001), and time factors (e.g., temporal discounting; Frederick et al., 2002). Hence, it may be informative to not only study to what extent the various choices representing the two perspectives are nominally identical (i.e., Research Question 1) but also whether the different choices tap into the various classes of risk-related psychological mechanisms to similar extents.

In Study 2 we thus addressed our second research question: To what extent do potential differences between researchers' and laypeople's perspectives imply differences in the psychological mechanisms they tap into? To this end, we prompted a large pool of people about their perceived relevance of seven classes of psychological mechanisms (Lob et al., 2025) in the choices representing the two perspectives.

Study 1: Comparing and Contextualizing Choices From Both Perspectives

We addressed our first research question with a natural language processing approach. Specifically, using high-dimensional semantic representations obtained from a sentence-embedding model, we first quantified the semantic similarity between all 163 risky choices in the inventories representing

researchers' and laypeople's perspectives. Second and upon conducting an empirical calibration study to identify a threshold of semantic similarity above which most people perceive two choices as identical, we determined the overlap of the choices within and particularly across the two inventories.

Moreover, to contextualize the choices from both perspectives, we used the same analytic approach to compare these choices with three additional resources on hazards and risk-related behaviors: big life decisions, global hazards, and health hazards. Specifically, Camilleri (2023) collected 49 big life decisions from a representative U.S. sample (e.g., whether to get married, move to a new city, or pursue religion/spirituality). Considering that such choices almost inherently contain an element of risk and uncertainty, they allow investigating the relatedness of the choices representing laypeople's perspective. Furthermore, experts from the World Economic Forum (WEF) have recently published an inventory of 34 of the most pressing systemic hazards in their annual Global Risks Report (e.g., misinformation and disinformation, pollution, lack of economic opportunity; World Economic Forum, 2024). This resource offers a broader context for examining how well the inventories representing researchers' and laypeople's perspectives align with global and systemic hazards recognized by experts. Finally, the GBD repository provides a comprehensive data set of 206 hazards for quantifiable health outcomes (e.g., chewing tobacco, cyclist road injuries; Murray et al., 2020). This resource can be used to assess the degree to which the inventories representing researchers' and laypeople's perspectives may particularly reflect health-related choices.

Method

Materials

To represent researchers' and laypeople's perspectives, we used the two above-described inventories of risky choices developed in precursory projects (see the Researchers' Perspective and Laypeople's Perspective sections). Specifically, researchers' perspective was represented by an inventory of 63 real-life risky choices (derived from frequency measures) typically used in the behavioral sciences (e.g., steal something, sunbathe without sunscreen). Laypeople's perspective was represented by an inventory of 100 choices reported by representative Swiss population samples

(e.g., quit one's job, use social media; Frey & Fischer, 2025). For comparability, choices in the inventory representing laypeople's perspective (originally worded in German) were translated into English, and choices from both inventories were normalized so that they all followed the same grammatical structure (i.e., "to" + [verb phrase]) and used second-person personal pronouns (i.e., "your" instead of "one's").

As outlined above, in a secondary analysis, we compared these choices with three external resources on choices and hazards, namely, big life decisions ($N = 49$ choices; Camilleri, 2023), hazards from the WEF's Global Risks Report ($N = 34$ hazards; World Economic Forum, 2024), and health-related hazards as identified by the GBD repository ($N = 206$ hazards; Murray et al., 2020).

Analysis Plan

Given the vast number of choice pairs to compare (i.e., 13,203, resulting from calculating the number of each possible pairwise combination of the 163 choices: $163 \times (163 - 1) \div 2$), relying on human judgments alone to determine the similarity of each pair was infeasible. Instead, we used natural language processing to scale up the comparison of choices. Specifically, to numerically represent the choices and hazards in the different inventories, we obtained their semantic embeddings using a transformer-based sentence encoder: the sentence-bidirectional encoder representations from transformers (SBERT) model (Reimers & Gurevych, 2019; for a recent application, see Bhatia, 2024) that we accessed via Hugging Face's application programming interface. In short, semantic embeddings are numerical representations of words, phrases, or sentences that capture their meaning in high-dimensional space. The SBERT model we used represents sentences as 768-dimensional vectors, where each dimension captures latent linguistic features.

Using the embeddings, we calculated cosine similarities as a measure of semantic similarity between choices and hazards across inventories. Cosine similarities are a common measure of similarity between vectors that captures the cosine of the angle between them. In the context of semantic embeddings, cosine similarities are used to quantify the similarity of the meanings of two words, phrases, or sentences (e.g., Bhatia, 2024; Hussain et al., 2024). Research on SBERT has shown a high correlation between cosine

similarities obtained from SBERT and semantic textual similarity scores obtained from human judgments (Reimers & Gurevych, 2019). Cosine similarities can range from -1 to 1 , with values close to -1 indicating opposing relationships (vectors with opposite directions), values close to 0 indicating no meaningful relationships (orthogonal vectors), and values close to 1 indicating high similarity (vectors with the same direction). In practice, however, cosine similarities computed from semantic embeddings rarely fall much below 0 because even words with opposite meanings (e.g., hot and cold) often appear in similar contexts (vor der Brück & Pouly, 2019). To illustrate, the sentences "The quick brown fox jumps over the lazy dog" and "The fast beige fox leaps over the sluggish labradoodle" have a cosine similarity of 0.81 . By contrast, the former sentence and "She sells seashells by the sea shore" have a cosine similarity of $.01$. Examples of choices and hazards entered into SBERT as well as the resulting semantic similarities are shown in Supplemental Table S1.

In addressing Research Question 1, we determined how many choices in the inventory representing researchers' perspective can be considered identical to choices in the inventory representing laypeople's perspective. We did so using choices' semantic similarity (as measured by their pairwise cosine similarity): Choices that had a semantic similarity above a certain threshold were considered identical; choices below this threshold were considered distinct. We determined this threshold using a signal detection analysis on human majority judgments of pairs of choices (see the Calibration Study section).

We used Python 3.10.12 (Python Software Foundation, 2023) to obtain the semantic embeddings and compute cosine similarities. All further analyses were performed using R Version 4.3.3 (R Core Team, 2024). The data, materials, and code associated with this project as well as the preregistration are publicly available on the Open Science Framework at <https://osf.io/f6e7q/>.

Calibration Study

To establish a threshold of semantic similarity above which humans perceive two choices to be the same, we conducted a calibration study. According to the self-evaluation checklist of the ethics committee of the Faculty of Arts and Social Sciences at the University of Zurich, this study did not require an in-depth review and further ethical approval.

Participants

The purpose of this calibration study was to establish a simple majority of human judgments on whether two choices are the same or not. We recruited 25 English-speaking participants located in the United States via Prolific, following previous findings on appropriate sample sizes when using the simple majority of participants' evaluations for analysis (Kuroda et al., 2023; Wolf et al., 2013). Specifically, these studies found that after approximately 25 participants, the incremental effect of additional participants on collective evaluations was negligible. Median study completion time was 8.2 min, and participants received a payment of £0.91 for their participation. Details on data collection and exclusion, sample composition, and quality control are included in [Supplemental Material I.a](#).

Materials and Procedure

Onboarding. Participants first read about the background and purpose of the study and provided informed consent. They then reported sociodemographic information (i.e., age, gender, level of education, employment status, and yearly income). To be permitted to continue the study, participants were informed that they had to turn off any VPN they were using and pass a Google reCAPTCHA bot check.

Instructions and Comprehension Check. Participants received the following information for the main task:

In the following section, we will present to you 100 pairs of phrases. For each pair, please indicate whether or not you consider these phrases to describe the same underlying choice. There is no right or wrong answer, we are just interested in your perception.

Please note that some phrases may appear more than once.

To make sure that participants had accurately understood the task, we then included a comprehension check ("To show that you have fully understood the instructions, please select the option that describes your next task."). This check consisted of eight possible responses shown in random order, only one of which was correct. Participants had two attempts to choose the correct answer. If they failed the comprehension check twice, they were not permitted to continue the study.

Similarity Ratings of Choice Pairs. In the main task, all participants were presented with the same predetermined random subset of 100 choice pairs (out of 13,203). To ensure that participants evaluated choice pairs with varying degrees of semantic similarity, we drew pairs from 10 equally spaced segments across the entire range of semantic similarities. One choice pair (i.e., comparing the choices "to demand a pay raise" and "to put yourself in danger at work") was removed post hoc because the choice "to put yourself in danger at work" had become defunct in the inventory representing laypeople's perspective (which was still in development during Study 1) due to a lower classification confidence in the inventory development procedure (for details, see Frey & Fischer, 2025). The choice pairs were presented in random order. For each choice pair (e.g., "to threaten someone" and "to give birth naturally," which have a cosine similarity of .1; "to not wear a seatbelt in a car" and "to drive a car without a seatbelt," which have a cosine similarity of .9), participants had to indicate whether they considered the two choices to be the same or not ("In your view, do these two phrases describe the same underlying choice?"). The distribution of the semantic similarities (i.e., in terms of cosine similarities) of these 99 pairs of the choices is displayed in Panel A of [Supplemental Figure S1](#), and the entire list of choice pairs that participants evaluated can be found on the Open Science Framework at <https://osf.io/f6e7q/>. We included three attention checks at random points throughout this task (e.g., "This is an attention check. Please select the option starting with No."). Participants had to pass at least two attention checks to complete the study and receive payment.

Focus During Study. At the end of the study, we asked participants to indicate their focus during the study on a slider scale from 0 (*not at all focused*) to 100 (*very focused*). We made clear to participants that their answer would not affect their payment:

To ensure the quality of our analyses, it is essential that we only include data from participants who were focused during the study. How focused were you during this study in general? (Don't worry, this will not affect your payment, you will receive the payment code either way.)

Identification of Threshold

Of the 99 choice pairs, 24 were classified as the same by a simple majority of participants (i.e., at least 13 out of 25). For example, "to not wear a

seatbelt in a car” and “to drive a car without a seatbelt” (cosine similarity = .9) were considered the same by most participants, whereas “to pollute the environment” and “to take out a loan” (cosine similarity = .18) were not. A receiver operating characteristic curve analysis based on these judgments and the corresponding semantic similarities revealed a threshold of .65 (see Panel B of Supplemental Figure S1), indicating that pairs with a similarity at or above this threshold were typically classified as the same.

Results

Semantic Similarity Between Choices

We first report descriptive statistics on the semantic similarities between choices: Within the two inventories representing researchers’ and laypeople’s perspectives, average semantic similarities were relatively low at approximately $M = 0.3$. This indicates that both inventories individually contain only little redundancy—which corroborates the preceding inventory development processes. The range of semantic similarity was slightly larger within the inventory representing laypeople’s perspective (range = $-.08, 1$) as compared with the inventory representing researchers’ perspective (range = $.01, .86$). More importantly, across the two inventories, the mean semantic similarity between choices was $M = 0.28$ (range = $-.04, 1$), indicating a similarly low level of redundancy between the two inventories as within each of them. Table 1 contains the average and range of similarities of all comparisons, and the complete distributions of semantic similarities within and across different inventories are illustrated in Supplemental Figure S2.

Panel A of Figure 1 illustrates these semantic similarities (i.e., in terms of cosine similarities of

the choices’ embeddings) between the various risky choices, within and across the two inventories. The blue and orange shapes represent the choices from the inventories representing researchers’ and laypeople’s perspective, respectively. The choices are arranged in a force-directed graph layout using the Fruchterman–Reingold algorithm, meaning choices that have a higher degree of semantic similarity to many other choices (e.g., “to try something new” from the inventory representing researchers’ perspective or “to leave the house” from the inventory representing laypeople’s perspective) organize as clusters in the center of the graph, whereas choices with fewer ties (e.g., “to express a dissenting opinion” from the researcher’s inventory or “to pay digitally” from the inventory representing laypeople’s perspective) are placed more marginally (Fruchterman & Reingold, 1991). The tone of the lines connecting choices represents the degree of semantic similarity, with darker lines indicating a higher degree of semantic similarity.

Overlap of Risky Choices Between Perspectives

To identify nominally identical choices, we next applied the empirically derived similarity threshold of .65 from our calibration study to all 13,203 unique pairs of choices. Retaining one randomly selected label from each pair at or above the threshold reduced the total number of choices from 163 to 103. Examples of choices subsumed under another label due to high semantic similarity include the following: “to have an affair” was subsumed under “to cheat on your partner” (inventory representing laypeople’s and researchers’ perspective, respectively; similarity = .76), “to insult someone” was subsumed under “to hurt someone’s feelings” (both from the inventory representing researchers’

Table 1

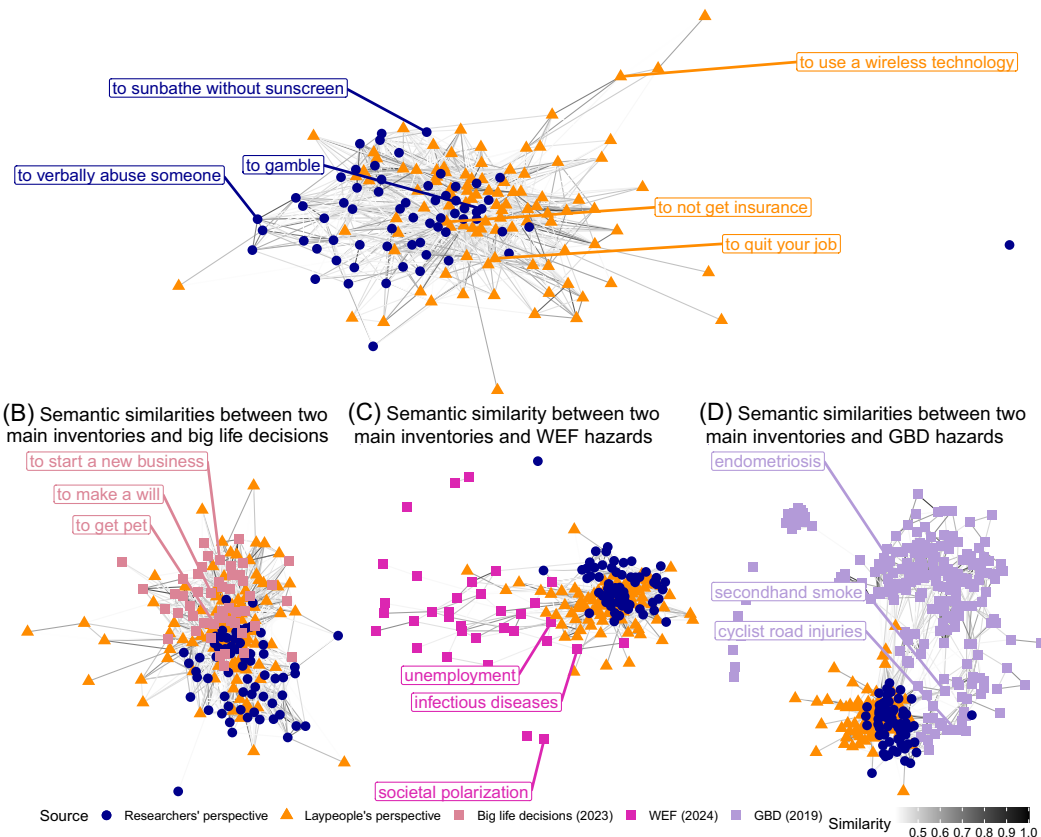
Average Semantic Similarity Between Inventories

Inventory	Res.	Lay.	BLD	WEF	GBD
Res.	0.33 [0.01, 0.86]	0.28 [−0.04, 1.00]	0.25 [−0.12, 0.84]	0.12 [−0.13, 0.63]	0.07 [−0.12, 0.75]
Lay.		0.30 [−0.08, 1.00]	0.28 [−0.08, 1.00]	0.11 [−0.12, 0.68]	0.05 [−0.13, 0.66]
BLD			0.32 [0.05, 0.90]	0.10 [−0.15, 0.61]	0.04 [−0.14, 0.72]
WEF				0.23 [−0.01, 0.77]	0.08 [−0.15, 0.75]
GBD					0.17 [−0.09, 0.93]

Note. Values in brackets represent the range of semantic similarity for the respective comparison. Res. = researchers’ perspective ($N = 63$); Lay. = laypeople’s perspective ($N = 100$); BLD = big life decisions ($N = 49$); WEF = World Economic Forum hazards ($N = 34$); GBD = Global Burden of Disease hazards ($N = 206$).

Figure 1
Semantic Similarities Between Risky Choices

(A) Semantic similarities between researchers' and laypeople's inventories



Note. Semantic similarities as measured by cosine similarities between the risky choices representing researchers' and laypeople's perspectives ($N = 163$) in Panel A, with big life decisions ($N = 163 + 49$) in Panel B, WEF hazards ($N = 163 + 34$) in Panel C, and health-related hazards in the GBD repository ($N = 163 + 206$) in Panel D. The shapes (i.e., choices or hazards) are arranged in a force-directed graph layout, where choices/hazards that have a higher degree of semantic similarity to many other choices/hazards are placed in the center of the graph, whereas choices/hazards with fewer ties are placed more marginally (for details on the algorithm used to produce the figure, see Fruchterman & Reingold, 1991). The tone of the lines connecting shapes represents the degree of semantic similarity: Darker lines indicate a higher degree of similarity. Similarities below .4 are not depicted. See Supplemental Figure S3 for a version of this figure that is colored by domain. WEF = World Economic Forum; GBD = Global Burden of Disease. See the online article for the color version of this figure.

perspective; similarity = .7), and "to paraglide," "to jump off a bridge," and "to bungee jump" were subsumed under "to skydive" (all from the inventory representing laypeople's perspective; similarities = .75, .76, and .8, respectively). Of the remaining 103 choices (see Figure 2), 30 (29%) were unique to the inventory representing the researchers' perspective (e.g., drive after drinking alcohol, sunbathe without sunscreen, spend money impulsively), 54 (52%) were unique to the inventory representing the laypeople's perspective (e.g.,

become self-employed, get divorced, use a smartphone), and 19 (18%) were shared by both inventories (e.g., drink alcohol, have unprotected sex, travel alone), representing the overlap between inventories. All 103 unique choices, including the 59 choices that were incorporated into the remaining 103 choices, are also listed in Supplemental Table S2, and the semantic relationships between this subset of choices, big life decisions, WEF global hazards, and GBD hazards are shown in Supplemental Figure S4.

Figure 2
Venn Diagram of 103 Choices



Note. These choices represent consolidated researchers' perspective ($N = 30$; blue), laypeople's perspective ($N = 54$; orange), and their intersection ($N = 19$). For brevity, choices are represented without the infinitive marker "to." See the online article for the color version of this figure.

Semantic Similarity as a Function of Life Domain

Examining the semantic similarity between choices in the two perspectives through the lens of life domains also reveals patterns beyond those of the inventories themselves. As shown in Panel A of Supplemental Figure S3, choices within the same life domain (i.e., financial, health, occupational, recreational, social, traffic; Frey & Fischer; Josef et al., 2016) tend to be grouped together, indicating some degree of within-domain semantic similarity. Recreational and traffic choices, in particular, group together relatively strongly irrespective of the inventory of origin, suggesting a higher semantic comparability of choices in the respective domains. There is somewhat more semantic variability in the other domains, most noticeably so in the social domain: Social choices from the researchers' inventory (many of which are of the more antisocial or delinquent flavor, e.g., to verbally abuse someone, to engage in an illegal activity) were placed apart from many social choices in the laypeople's inventory (e.g., to address a person, to have children, to use a smartphone).

Contextualizing Risky Choices From Both Perspectives

To compare the conceptualizations of risky choices from both perspectives to three external resources (49 big life decisions, 34 WEF hazards, and 206 GBD hazards; Camilleri, 2023; Murray et al., 2020; World Economic Forum, 2024), we used the same method of quantifying semantic similarity between choices. Panels B, C, and D of Figure 1 illustrate the semantic similarities between choices in the two inventories and the three external resources. The range and average semantic similarity of big life decisions and choices in either inventory representing researchers' or laypeople's perspective were at a similar level as the average semantic similarity between choices in the two inventories (see Table 1), indicating an overall similar level of relatedness between those choices to big life decisions. By contrast, the average semantic similarity between choices in the two inventories compared with both WEF hazards and GBD hazards was substantially lower, suggesting a greater disjoint between either inventory and these more systemic (WEF) and health-related (GBD) hazards.

The placement of choices and hazards in Panels B–D of Supplemental Figure S3, relative to the domains of choices in the researchers' and laypeople's inventories, provides further insights into their semantic structure: Big life decisions were more closely grouped with financial and occupational choices as well as some social choices, reflecting the finance, career, and education categories as well as the family and relationships categories in the former (Camilleri, 2023). By contrast, big life decisions showed few semantic associations with traffic choices and many of the more delinquent social choices in the researchers' inventory. While a few GBD hazards were perhaps unsurprisingly grouped with health (semantically linking substance use to drug use disorders), traffic (linking behavior on the road with related injuries), and some social choices predominantly from the researchers' inventory (linking antisocial or violent risky choices with GBD hazards related to physical harm), identifying patterns for WEF hazards was not possible as most hazards were semantically far removed from choices in the researchers' or laypeople's inventory.

Discussion

Study 1 revealed a substantial discrepancy in the real-life risky choices typically investigated via frequency measures in the behavioral sciences and those reported by laypeople. Specifically, after identifying 103 unique choices from an initial set of 163, only 19 choices (18%) were present in both the inventories representing the researchers' and laypeople's perspectives. This indicates that a vast majority of choices were unique to either the inventory representing the researchers' or laypeople's perspective, suggesting that the current status quo of real-life choices studied in the form of frequency measures may not fully capture the diversity of choices encountered in everyday life.

Many choices representing researchers' perspective originate from clinical or developmental psychology, often focusing on potentially pathological or delinquent behaviors. This is reflected in the destructive nature of many choices representing researchers' perspective (e.g., destroy property, drive after drinking alcohol, spend money impulsively) and may point to a broader tendency in the behavioral sciences to focus on negative aspects of risk taking (Duell & Steinberg, 2020; Schonberg et al., 2011). Such choices also typically have a dominant, normative, or objectively better option (e.g., it is clearly better not to drive after drinking alcohol). By

contrast, the inventory representing laypeople's perspective includes choices that appear more balanced, involving both positive and negative outcomes (e.g., accept a new challenge at work, get surgery, use social media).

One interpretation of researchers' strong focus on pathological or delinquent behaviors could consist of the aim to optimize an instrument's measurement sensitivity: To identify people with particularly heightened risk-taking tendencies, it is important to examine choices that not only have potentially high consequences but also variability across individuals—which may be the case in many choices of the researchers' perspective (e.g., whether to destroy property). Although many choices of the laypeople's perspective also fulfill these criteria, some of the particularly salient and consequential choices that could also be considered big life decisions (e.g., whether to quit one's job, undergo chemotherapy, or carry out a pregnancy to term; Camilleri, 2023) may be too rare for diagnostic applications (floor effects). Conversely, other choices (e.g., to use a smartphone) may be too common (ceiling effects). That is, whereas these risky choices may not be ideal for diagnostic applications, behavioral research may still benefit from studying the diverse and at times novel risky choices (e.g., using a smartphone, using social media, or polluting the environment) to reveal how these choices unfold in different contexts, engage different psychological mechanisms, or affect different populations compared with the more established choices.

Possibly, the 19 choices that overlap between the two perspectives may represent the “best of both worlds”: Many of these are widespread enough to be relevant to many people while avoiding floor or ceiling effects. As such, they seem to meet the criterion of being able to differentiate between individuals. Moreover, many commonly implemented scales already include these choices. This means that some of the more frequently studied choices were also prevalent in the inventory representing laypeople's perspective. Although these 19 overlapping choices represent only a small subset of the total (18%), they indicate that the two inventories are not entirely disjoint.

Finally, we also compared the risky choices with external resources with different conceptualizations of hazards and choice. These analyses provide further guidance to better understand the qualitative nature of the non-overlapping choices of the two perspectives

(e.g., Supplemental Figure S4 suggests that relatively more of the choices of the laypeople's perspective are related to big life decisions; Camilleri, 2023). In this context, we also found that inventories containing choices (i.e., inventory representing researchers' perspective, inventory representing laypeople's perspective, and big life decisions) were more similar to each other than to the WEF hazards and health-related hazards from the GBD repository. This is in part due to the fact that all choices had the same grammatical structure and were worded as verb phrases (i.e., “to + [verb phrase]”) to which an embedding model like SBERT is particularly sensitive. In addition, however, the lower degree of semantic similarity between choices and hazards also indicates that at least some of the WEF (e.g., “nonweather-related natural disasters”) or GBD (e.g., “chronic lymphoid leukemia”) hazards are not closely related to the individual choices (i.e., in the sense that they do not frequently co-occur) in the text data SBERT is familiar with.

Study 2: Analyzing Which Classes of Psychological Mechanisms the Risky Choices From Both Perspectives May Tap Into

Despite the nominal divergence between perspectives, findings based on the choices in the researchers' perspective could still generalize to choices in the laypeople's perspective if they tap the same psychological mechanisms. By contrast, if this is not the case, measurement instruments calibrated on extant frequency measures may not fully capture or predict many real-life choices comprised in the laypeople's perspective. But what may be the psychological mechanisms at play when people make these choices?

Whereas traitlike predictors of decision making such as risk preference (e.g., Blais & Weber, 2006; Frey et al., 2017; Josef et al., 2016) or numeracy (e.g., Cokely et al., 2018) permit making mean-level predictions of (risky) choices, various state-specific psychological mechanisms may actually shape how people make risky choices in different situations and contexts. As outlined in the introduction, a recent taxonomy identified 83 such state-specific, psychological mechanisms, and these mechanisms are organized in seven classes of psychological functioning (Lob et al., 2025): mechanisms related to the subjective evaluation and perception of choice attributes (e.g., subjective

perception of values; Bernoulli, 1738; von Neumann & Morgenstern, 1944), time factors (e.g., time pressure; Busemeyer & Townsend, 1993), goals and motivation (e.g., specific needs; Mishra et al., 2017), cognitive resources and control (e.g., memory processes; Johnson et al., 2007), social factors (e.g., social norms; Gerrard et al., 2008), experience and knowledge (e.g., previous exposure; Loewenstein et al., 2001), and affect (e.g., regret; Loomes & Sugden, 1982). In Study 2, we built on the seven classes of this taxonomy to thus analyze the extent to which the various risky choices of the two perspectives tap classes of psychological mechanisms to a similar extent—or, conversely, whether there exist systematic discrepancies.

Method

Study 2 was reviewed and approved by the ethics committee of the Faculty of Arts and Social Sciences at the University of Zurich (#23.04.12).

Participants

We collected data from 825 English-speaking participants located in the United States via Prolific (sample size was determined by a prior simulation study; see <https://osf.io/f6e7q/>). Median study completion time was 14.6 min, and participants received a payment of £1.50 for their participation. Details on data collection and exclusion, sample composition, and quality control are described in [Supplemental Material I.b](#).

Materials and Procedure

Study 2 employed a 7 (choice blocks with 15 choices each; between) $\times$ 7 (mechanisms; within) mixed design. Predetermined blocks contained 15 random choices representing researchers' and laypeople's perspectives (including their overlap). Participants were randomly assigned to a block and rated all choices within this block such that each choice was rated by 116 to 119 participants. That is, a total of 105 choices was rated across all participants, with two choices (i.e., "to put yourself in danger at work" and "have a job with an insecure income") removed post hoc due to defunct choices in the inventory representing laypeople's perspective (which was still in development during Study 2; see [Supplemental Material IV](#)). That is, in total, 103 choices were

analyzed. The order of choices within a block was randomized.

Onboarding. As in the calibration study, participants first read about the background and purpose of the study and provided informed consent. They then reported sociodemographic information (i.e., age, gender, level of education, employment status, and yearly income). Participants were informed that they had to turn off any VPN they were using and pass a Google reCAPTCHA bot check to continue.

Instructions and Comprehension Check. Participants received the following information on the main task:

When people make choices, they may take various factors into account. In the next section, you will see a series of 15 different choices. We are interested in understanding the extent to which 7 factors would play a role in your decision-making process if you were to make these choices.

As in the calibration study, participants had two attempts to pass the comprehension check before being prevented from completing the study.

Perceived Relevance of Classes of Psychological Mechanisms. In the main task, participants rated the perceived relevance of seven classes of psychological mechanisms (Lob et al., 2025) for choices in different inventories on a continuous slider from 0 (*not at all*) to 100 (*very much*):

Imagine you are in a specific situation and about to decide whether or not to [risky choice]. When making this choice, to what extent would the following play a role for you?

With this phrasing and the respective operationalizations of the classes of psychological mechanisms (see [Table 2](#)), we aimed to capture and translate the core aspects of each class into everyday language that was understandable for laypeople. The wording was intentionally broad and inclusive to capture any signal of participants' consideration of the overarching classes of psychological mechanisms rather than particular mechanisms within each class. The decision to focus on the overarching classes was partly pragmatic given the large number of mechanisms in the taxonomy (83 in total) but also promised to capture a wider range of aspects that people may consider in their decision-making process.

We included six attention checks at random points throughout this task (e.g., "To show that you are still paying attention, please move the slider all the way to the left side of the scale."). Participants had to pass at least five attention checks to complete the study and receive payment.

Table 2
Classes of Psychological Mechanisms

Mechanisms related to the subjective evaluation and perception of	Operationalization
Choice attributes	“How appealing the choice options are.”
Time factors	“How much time you have to make a decision or how soon the outcomes will occur.”
Goals and motivation	“How strongly you want or need to pursue a goal.”
Cognitive resources and control	“How capable you are to focus your attention or control your actions.”
Social factors	“How other people behave or how they perceive you.”
Experience and knowledge	“How experienced or knowledgeable you are.”
Affect	“How you are feeling or expecting to feel.”

Focus During Study. As in the calibration study, participants finally indicated their focus during the study on a scale of 0 (*not at all focused*) to 100 (*very focused*).

Analysis Plan

We addressed Research Question 2 by comparing the perceived relevance of different classes of psychological mechanisms across choices in the two inventories (including their overlap). Specifically, we quantified these differences in mean-level ratings with Bayesian mixed effects models (see Oberauer, 2022) using the brms package in R (Version 2.21.0; Bürkner, 2017). The most complex (i.e., benchmark) model was specified as follows:

$$R_{pc} = \beta_0 + u_{0p} + u_{0c} + \beta_{1i}I_c + (\beta_{2m} + u_{1p} + u_{1c})M_c + \beta_{3im}(I_c \times M_c) + e_{pc}, \quad (1)$$

where R_{pc} is the perceived relevance of a class of mechanism for a choice c by a participant p , I_c is a dummy-coded variable for the inventory of the choice (researchers’, laypeople’s, or their overlap), M_c is a dummy-coded variable for the class of mechanism, and $I_c \times M_c$ is their interaction. The fixed effects β_0 to β_{3im} represent the intercept (β_0) and mean effects of each inventory (β_{1i}), class of mechanism (β_{2m}), and their interaction (β_{3im}). The random effects u_{0p} and u_{0c} are by-participant and by-choice random intercepts, respectively, and u_{1p} and u_{1c} are by-participant and by-choice random slopes for class of mechanism. e_{pc} is the residual error term. A beta distribution was used for the likelihood (Smithson & Verkuilen, 2006). We employed weakly informative priors for both the fixed and random effects. Further information

on model specification using a beta likelihood is included in Supplemental Material I.b, and the fitting procedure is detailed in the preregistration at <https://osf.io/f6e7q/>.

In defining this benchmark model, we used a maximal approach, aiming to include all random effects justified by the study design. Previous work has demonstrated the importance of including random effects for modeling hierarchically structured data, as omitting them can bias parameter estimates and increase the risk of false positives (Barr et al., 2013; Oberauer, 2022). However, in a simulation study conducted for the preregistration (see <https://osf.io/f6e7q/>), the maximal model failed to compile (likely due to the block randomization in the study design leading to many missing values). As a result, the benchmark model does not include random slopes for inventory or the interaction between inventory and class of mechanism. Nonetheless, including all classes of mechanisms and inventories in one model allowed us to directly compare the perceived relevance of each class of mechanism for choices in each inventory and thereby directly address Research Question 2.

To identify the model most strongly supported by the data while balancing goodness of fit and parsimony (Lee & Wagenmakers, 2014; Vandekerckhove et al., 2015), we sequentially compared the benchmark model with models with reduced random and fixed effects structures, using Bayes factors (BF) to guide model simplification (Wetzels et al., 2011). BFs were estimated using the Savage–Dickey density ratio method (Wagenmakers et al., 2010) with the R package bayestestR (Version 0.13.2; Makowski et al., 2019). The results of the model comparisons are reported in Supplemental Material II.c.

Finally, we examined the posterior distributions of estimated parameters to assess differences in

participants’ relevance ratings between inventories and classes of mechanisms. To evaluate whether these differences credibly deviated from zero, we obtained BF_s by comparing the posterior distributions with zero (Wetzels et al., 2011).

Results

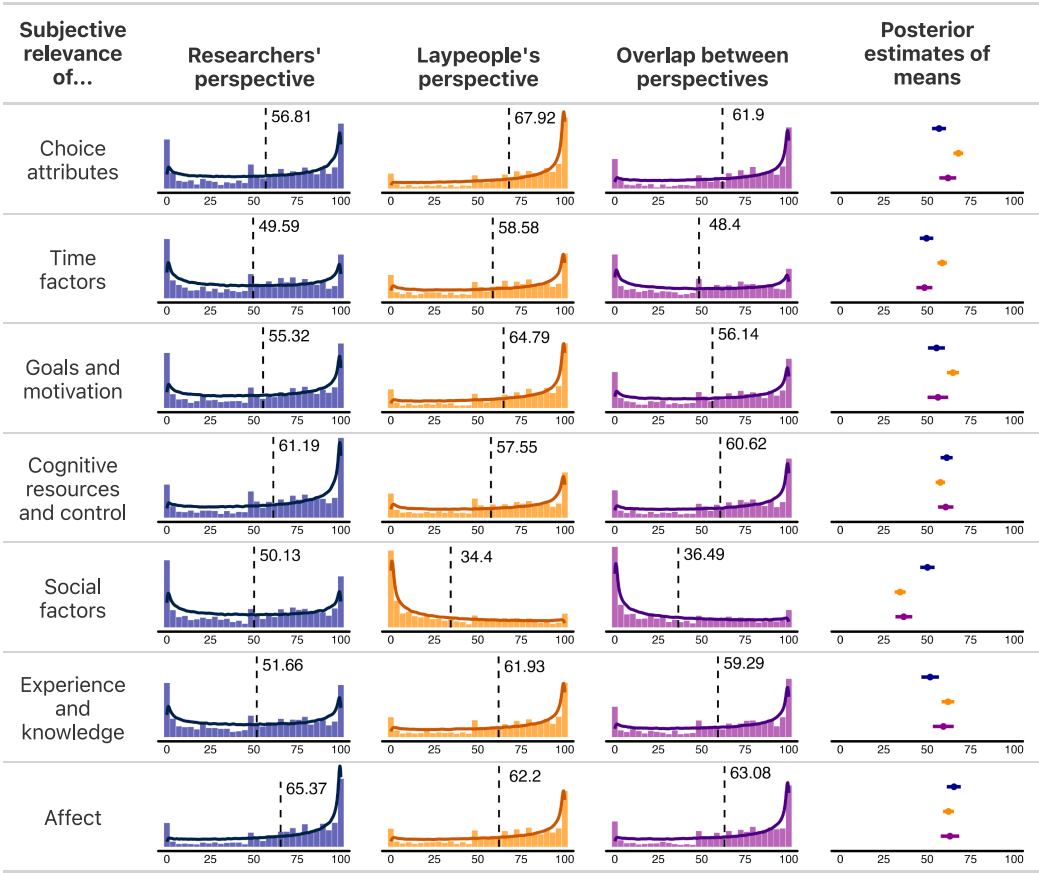
Perceived Relevance of Psychological Mechanisms

Participants’ ratings revealed distinct patterns in the perceived relevance of psychological mechanisms depending on whether a particular choice

was part of the inventory representing researchers’ perspective, laypeople’s perspective, or the overlap between the two. Figure 3 illustrates the empirical distributions of participants’ relevance ratings for each class of psychological mechanism and for two different inventories and their overlap, as well as the posterior estimates of the respective mean ratings (derived from the benchmark model specified in Equation 1).

On average, the classes of psychological mechanisms perceived to be most relevant were affect for choices representing researchers’ perspective ($M = 65.37$; 95% highest density interval [HDI] = 61.34, 69.36), choice attributes for choices

Figure 3
Rated Relevance of Classes of Psychological Mechanisms



Note. Columns 1–3: Empirical distributions of the rated relevance of psychological mechanisms, separately for the risky choices representing researchers’ perspective, laypeople’s perspective, and the choices in the overlap between the two. The vertical dashed lines in the histograms depict the estimated means, and each distribution is overlaid with the posterior predictives that visualize model fit. Column 4: Mean and 95% highest density intervals of the estimated mean relevance of the classes of mechanisms, separately for the two inventories and their overlap. See the online article for the color version of this figure.

representing laypeople's perspective ($M = 67.92$; $HDI = 65.07, 70.71$), and again affect for choices that were present in both perspectives ($M = 63.08$; $HDI = 57.66, 68.36$). Conversely, the classes of psychological mechanisms perceived to be the least relevant were time factors for choices representing researchers' perspective ($M = 49.59$; $HDI = 45.76, 53.49$), social factors for choices representing laypeople's perspective ($M = 34.4$; $HDI = 31.28, 37.60$), and again social factors for choices that were present in both perspectives ($M = 36.49$; $HDI = 31.61, 41.31$). As indicated by the standard deviations of these ratings, the perceived relevance of classes of psychological mechanisms varied greatly. The distributions of participants' ratings for individual choices are shown in [Supplemental Figures S5–S11](#).

Comparison of Perspectives

The benchmark model specified in [Equation 1](#), which received the strongest support from the data (see [Supplemental Material II.c](#)), revealed credible differences in the perceived relevance of five out of seven classes of psychological mechanisms. [Supplemental Table S3](#) contains the effect sizes and BFs for these comparisons of perceived relevance across different inventories and classes of mechanisms.

Specifically and using the common naming convention to qualify the strength of evidence in the form of BFs ([Jeffreys, 1961](#); [Wetzels et al., 2011](#)), there was decisive evidence that participants perceived the relevance of some classes of mechanisms differently for the two inventories (see the right-most column of [Figure 3](#) for the posterior estimates of the means), namely, for choice attributes (higher perceived relevance for choices representing laypeople's perspective; $BF_{10} = 2.16 \times 10^3$), time factors (higher in choices representing laypeople's perspective; $BF_{10} = 3.81 \times 10^2$), and social factors (higher in choices representing researchers' perspective; $BF_{10} = 6.67 \times 10^5$). There was also very strong evidence for differences in the perceived relevance of experience and knowledge (higher in choices representing laypeople's perspective; $BF_{10} = 4.86 \times 10^1$) as well as goals and motivation (higher in choices representing laypeople's perspective; $BF_{10} = 3.41 \times 10^1$). By contrast, there was anecdotal evidence that participants did not perceive the relevance of cognitive resources and control ($BF_{10} = 6.05 \times 10^{-1}$) and affect ($BF_{10} = 3.69 \times 10^{-1}$) differently for the

two inventories. Additional model parameters are reported in [Supplemental Table S4](#).

When comparing choices representing researchers' perspectives with choices in the overlap between perspectives, there was decisive evidence for a difference in the perceived relevance of social factors (higher in choices representing researchers' perspective; $BF_{10} = 1.28 \times 10^3$) and anecdotal evidence for a difference in the perceived relevance of experience and knowledge (higher in the overlap between inventories; $BF_{10} = 1.84$). Comparing choices representing laypeople's perspective with choices in the overlap between inventories, there was decisive evidence for a difference in the perceived relevance of time factors (higher in choices representing laypeople's perspective; $BF_{10} = 2.17 \times 10^2$), substantial evidence for differences in the perceived relevance of goals and motivation (higher in choices representing laypeople's perspective; $BF_{10} = 5.47$), and anecdotal evidence for differences in the perceived relevance of choice attributes (higher in choices representing laypeople's perspective; $BF_{10} = 1.95$).

Discussion

Despite the small overlap in the choices representing researchers' and laypeople's perspectives (see Study 1), Study 2 suggests that there were at least some similarities between perspectives in the perceived relevance of classes of psychological mechanisms. Specifically, across the two perspectives and their overlap, participants perceived cognitive resources and control, as well as affect, to be similarly relevant when making risky choices.

However, these similarities between perspectives represent the exception rather than the rule: For the other five classes of psychological mechanisms, participants perceived differing levels of relevance for the choices comprised in the two perspectives. Four of these five classes of mechanisms—choice attributes, time factors, goals and motivation, and experience and knowledge—were on average perceived as more relevant in the inventory representing laypeople's perspective than in the inventory representing researchers' perspective. Overall, the most pronounced difference between the inventories can be found in the perceived relevance of social factors (see [Supplemental Table S3](#)): Social factors were rated as more important for choices representing researchers' perspective than for choices representing laypeople's perspective. This may reflect the high

occurrence of the social domain in choices representing researchers' perspective and suggests that these choices may, at least in participants' views, be more susceptible to factors such as peer influence (Irwin & Millstein, 1986), social norms (Gerrard et al., 2008), or accountability (Tetlock, 1992).

Overall, Study 2 suggests that participants on average perceive the various classes of psychological mechanisms to have a different relevance to the choices of the two inventories (and their overlap). These findings further highlight the potential issue identified in Study 1: Specifically, the two perspectives seem to differ not only at the level of choices (i.e., in terms of whether they involve the nominally identical risky choices) but also at the level of psychological mechanisms the respective risky choices may tap into. These discrepancies raise an important issue for decision-making research: Focusing primarily on extant frequency measures of risky choices to develop and calibrate various measurement instruments (i.e., behavioral tasks and propensity measures) may not accurately capture the relevance of certain mechanisms in other real-world choices as reported by laypeople. This, in turn, may have important implications for the field's ability to understand and predict many real-world choices.

General Discussion

A key objective of the behavioral sciences is to understand and predict how people ultimately make risky choices in life. In this article, we aimed to address two fundamentally important questions in this regard: First, to what extent are the real-life choices tapped by extant frequency measures nominally the same as those that people report to face? Second, to what extent do potential differences between these perspectives imply differences in the tapped psychological mechanisms?

The present findings reveal substantial discrepancies between the risky choices investigated by behavioral scientists via frequency measures and those laypeople reported to actually face in life—both at the level of the nominal choices themselves and the psychological mechanisms presumably involved in the respective risky choices. That is, although several choices in the inventory representing researchers' perspective (e.g., whether to steal, get into a fight, or destroy property) can

help researchers better understand and potentially prevent delinquent behavior, people face a broader range of risky choices that have received comparatively less attention to date. This gap may affect the extent to which widely used measurement instruments (i.e., behavioral tasks or questionnaires of people's risk taking) generalize to real-life decision-making contexts because such measurement instruments are often calibrated with specific frequency measures of risky choices thought to occur in life.

The dominant use of the extant frequency measures may in parts be related to the aim of optimizing on diagnostic success (see the Discussion section of Study 1) but at least to some extent also to the dynamics of scientific practice: Once certain frequency measures became established, researchers were perhaps more likely to use these measures again in the future. This reinforced their prominence in the literature and their position as appropriate benchmarks for measurement validation (Hull, 1988; Kuhn, 1962), contributing to a narrower focus on a subset of risky choices. Institutional endorsement may have further contributed to this self-reinforcing cycle: For instance, the Alcohol Use Disorders Identification Test (Saunders et al., 1993) is recommended by the World Health Organization for the identification of substance abuse, which likely facilitated its widespread adoption and its use for calibrating other measurement instruments such as the balloon analogue risk task (Lejuez et al., 2002).

To be clear and as outlined in the introduction, there is likely no definitive answer or ground truth to the question of what constitutes relevant risky choices. Moreover, the results from this article should not be taken as evidence for a general absence of research on real-life risky choices that are exclusively represented in laypeople's perspective: We focused on risky choices tapped by frequency measures because these are typically used to assess individual differences, yet this is naturally not the only approach to assessing people's real-life decision making (Gigerenzer, 2006; Klein, 2017; Siegrist & Cvetkovich, 2000; Slovic, 1987). That said, by addressing the two overarching research questions of this article, we have aimed to uncover commonalities and discrepancies between two major perspectives on what constitutes relevant real-life risky choices, as well as to contextualize these perspectives within three external resources on important choices and hazards more generally. Combining

the conclusions of both studies, the behavioral sciences may have little to lose and potentially much to gain by “casting a wider net”—by including a larger set of choices when calibrating and assessing the real-life relevance of various measurement instruments.

Interestingly, many risky choices reported by laypeople or studied via frequency measures by researchers may not qualify as decisions under risk in the narrower sense but rather as decisions under uncertainty: In the former, the probabilities of specific outcomes are explicitly known (e.g., playing roulette), whereas in the latter, it is at least partially unknown what outcomes might even occur, or with which probabilities specific outcomes can occur (e.g., founding a startup; see also Footnote 1). A similar pattern is also reflected in many measurement instruments of the behavioral sciences that are portrayed as measures of risk taking or that even carry the term *risk* in their name—such as the balloon analogue risk task or the Domain-Specific Risk-Taking Scale—even though they do not involve decisions under risk in the sense of decisions with known probabilities of specific outcomes. Vice versa, some studies focusing on behavioral tasks that explicitly implement decisions under risk (e.g., paired-lottery choices; Holt & Laury, 2002) have validated these tasks against real-life risky choices mostly entailing decisions under uncertainty (e.g., Anderson & Mellor, 2008). Overall, this illustrates that both laypeople and researchers often appear to use risky choice in a relatively inclusive manner, that is, as an umbrella term to describe various decision-making situations.

Constraints on Generality

Our conclusions should be interpreted in light of some constraints, two of which we will address in detail. First, our conclusions regarding the overlap of risky choices in the inventories representing researchers’ and laypeople’s perspectives naturally rest on particular samples of risky choices. Although these samples were large—that is, the laypeople’s perspective is based on choices reported by representative Swiss samples comprising more than 4,000 individuals (Frey & Fischer, 2025), and the inventory representing researchers’ perspective was distilled from an initial list of more than 600 frequency measures from various subfields within the behavioral sciences—we cannot entirely rule out the possibility that some relevant choices were not

captured by either approach. Moreover, while the inventory representing the laypeople’s perspective may adequately describe the ecology of risk in other Western societies, it may not capture important risky choices faced by people in other parts of the world.

Second, our conclusions about the relevance of psychological mechanisms are based on participants’ perceptions of how relevant the various classes of mechanisms would be if they were about to make these choices. Judging the relevance of psychological mechanisms may be more difficult for some mechanisms (e.g., mechanisms operating automatically or without conscious awareness, such as the class of mechanisms described by cognitive resources and control) than others (see Ericsson & Simon, 1998; Nisbett & Wilson, 1977, for a long-standing debate about individuals’ ability to provide insights into internal processes). Therefore, the findings from this study primarily speak to the perceived relevance of various classes of psychological mechanisms for different choices. Although this distinction may limit the generality of our findings, in our view the discrepancies in perceived relevance of different classes of psychological mechanisms between the two perspectives highlight which mechanisms warrant further investigation (see below).

Opportunities for Future Research

In what follows we make four recommendations for future research. First, we recommend that the inventory representing laypeople’s perspective is used as a basis for the development of new frequency measures. The selection of choices for new measurement instruments should be guided both by methodological considerations (e.g., diagnostic success) and theoretical interests in particular psychological mechanisms. For the latter, the ratings of perceived relevance in Study 2 may constitute a valuable starting point. Novel (scales of) frequency measures would allow yet other researchers to calibrate various measurement instruments on a broader set of real-life choices.

Second, future research could leverage domain-specific patterns when selecting risky choices for the development of new frequency measures. Prior work has shown that risk preference varies systematically across domains such as health, finance, or social life (e.g., Weber et al., 2002), a finding recently corroborated by language-based analyses showing that risky behaviors within the

same domain tended to share similar linguistic representations (Bhatia, 2024). A similar domain-specific pattern was apparent in the semantic similarities of choices in the researchers' and laypeople's inventories (see Supplemental Figure S3): Choices belonging to the same life domain tended to have a higher degree of semantic similarity, almost irrespective of inventory. This suggests a concrete avenue for future research: Existing measurement instruments could be tested for generalizability against choices in the laypeople's inventory that are semantically similar to or distinct from established frequency measures within the same domain. This would allow researchers to identify domains in which existing measurement instruments already capture relevant psychological processes and those in which additional frequency measures may be most urgently needed.

Third, although the present research has provided insights into the perceived relevance of different classes of psychological mechanisms, a more fine-grained analysis of individual psychological mechanisms—rather than their overarching classes—may be particularly insightful. For instance, researchers could investigate whether the higher perceived relevance of time factors in choices from the inventory representing laypeople's perspective (as compared with the inventory representing researchers' perspective) is primarily driven by perceived time pressure (Busemeyer & Townsend, 1993), temporal discounting (Samuelson, 1937), or the longevity of consequences (Slovic, 1987).

Fourth, researchers may further study the various risky choices of the two inventories (or their overlap) in contexts that carry real consequences. To this end, techniques such as ecological momentary assessments may constitute a fruitful tool for real-life applications. Building on seminal work in this area that investigated everyday risk perception (Hogarth et al., 2007; Hogarth et al., 2011) and studies of high-stakes decisions in applied settings (Klein, 2017), future research could use this methodology to simultaneously obtain a detailed representation of the risky choices at play in people's lives and learn about psychological mechanisms involved at different steps of the decision-making process. Such efforts could be complemented by aggregate statistical or actuarial data on risky behaviors or their consequences (e.g., Gigerenzer, 2006). Although the latter may not necessarily be suitable for assessing individual

differences, they could provide a valuable benchmark for assessing the prevalence of such choices.

Conclusion

The behavioral sciences and research on judgment and decision making in particular aim to better understand and predict how people make decisions under risk and uncertainty. This requires the research conducted in these fields to generalize from the lab—often as assessed via measurement instruments such as behavioral tasks or propensity measures—to real life. This can be achieved if the respective measurement instruments are calibrated on specific frequency measures of risky choices that tap the same choices as people actually face in their lives—or if the psychological mechanisms at play in these choices are at least largely the same. In two studies, we have shown that although some of the rather widespread frequency measures tap risky choices that people also report to encounter in their lives, neither of these requirements is met entirely: Many risky choices commonly assessed via frequency measures differ from the choices individuals encounter, and they seem to tap into different classes of psychological mechanisms. To enhance the applicability and generalizability of research findings, research would benefit from calibrating measurement instruments on a broader set of real-life choices. Doing so may contribute to a deeper understanding of decision-making processes in everyday life.

References

- Aklin, W. M., Lejuez, C. W., Zvolensky, M. J., Kahler, C. W., & Gwadz, M. (2005). Evaluation of behavioral measures of risk taking propensity with inner city adolescents. *Behaviour Research and Therapy*, 43(2), 215–228. <https://doi.org/10.1016/j.brat.2003.12.007>
- Allen, J. P., Reinert, D. F., & Volk, R. J. (2001). The Alcohol Use Disorders Identification Test: An aid to recognition of alcohol problems in primary care patients. *Preventive Medicine*, 33(5), 428–433. <https://doi.org/10.1006/pmed.2001.0910>
- Anderson, L. R., & Mellor, J. M. (2008). Predicting health behaviors with an experimental measure of risk preference. *Journal of Health Economics*, 27(5), 1260–1274. <https://doi.org/10.1016/j.jhealeco.2008.05.011>
- Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and*

- Language*, 68(3), 255–278. <https://doi.org/10.1016/j.jml.2012.11.001>
- Bernoulli, D. (1738). Exposition of a new theory on the measurement of risk. *Econometrica*, 22(1), 22–36. <https://doi.org/10.2307/1909829>
- Bhatia, S. (2024). Exploring variability in risk taking with large language models. *Journal of Experimental Psychology: General*, 153(7), 1838–1860. <https://doi.org/10.1037/xge0001607>
- Blais, A.-R., & Weber, E. U. (2006). A Domain-Specific Risk-Taking (DOSPERT) Scale for adult populations. *Judgment and Decision Making*, 1(1), 33–47. <https://doi.org/10.1017/S1930297500000334>
- Buelow, M. T., & Suhr, J. A. (2009). Construct validity of the Iowa Gambling Task. *Neuropsychology Review*, 19(1), 102–114. <https://doi.org/10.1007/s11065-009-9083-4>
- Bürkner, P.-C. (2017). Brms: An R package for Bayesian multilevel models using stan. *Journal of Statistical Software*, 80, 1–28. <https://doi.org/10.18637/jss.v080.i01>
- Bussemeyer, J. R., & Townsend, J. T. (1993). Decision field theory: A dynamic-cognitive approach to decision making in an uncertain environment. *Psychological Review*, 100(3), 432–459. <https://doi.org/10.1037/0033-295X.100.3.432>
- Camilleri, A. R. (2023). An investigation of big life decisions. *Judgment and Decision Making*, 18, Article e32. <https://doi.org/10.1017/jdm.2023.30>
- Clark, W. A. V., & Lisowski, W. (2017). Prospect theory and the decision to move or stay. *Proceedings of the National Academy of Sciences*, 114(36), E7432–E7440. <https://doi.org/10.1073/pnas.1708505114>
- Cokely, E. T., Feltz, A., Ghazal, S., Allan, J. N., Petrova, D., & Garcia-Retamero, R. (2018). Skilled decision theory: From intelligence to numeracy and expertise. In K. A. Ericsson, R. R. Hoffman, A. Kozbelt, & A. M. Williams (Eds.), *The Cambridge handbook of expertise and expert performance* (2nd ed., pp. 476–505). Cambridge University Press.
- Duell, N., & Steinberg, L. (2019). Positive risk taking in adolescence. *Child Development Perspectives*, 13(1), 48–52. <https://doi.org/10.1111/cdep.12310>
- Duell, N., & Steinberg, L. (2020). Differential correlates of positive and negative risk taking in adolescence. *Journal of Youth and Adolescence*, 49(6), 1162–1178. <https://doi.org/10.1007/s10964-020-01237-7>
- Edwards, W. (1954). The theory of decision making. *Psychological Bulletin*, 51(4), 380–417. <https://doi.org/10.1037/h0053870>
- Ericsson, K. A., & Simon, H. A. (1998). How to study thinking in everyday life: Contrasting think-aloud protocols with descriptions and explanations of thinking. *Mind, Culture, and Activity*, 5 (3), 178–186. https://doi.org/10.1207/s15327884mca0503_3
- Fairley, K., Parelman, J. M., Jones, M., & Carter, R. M. (2019). Risky health choices and the balloon economic risk protocol. *Journal of Economic Psychology*, 73, 15–33. <https://doi.org/10.1016/j.joep.2019.04.005>
- Fox, C. R., & Ülkümen, G. (2011). Distinguishing two dimensions of uncertainty. In W. Brun, G. Kirkebøen, & H. Montgomery (Eds.), *Essays in judgment and decision making* (pp. 21–35). Universitetsforlaget. <https://www.ssrn.com/abstract=3695311>
- Frederick, S., Loewenstein, G., & O'Donoghue, T. (2002). Time discounting and time preference: A critical review. *Journal of Economic Literature*, 40(2), 351–401. <https://doi.org/10.1257/002205102320161311>
- Frey, R. (2021). Psychological drivers of individual differences in risk perception: A systematic case study focusing on 5G. *Psychological Science*, 32(10), 1592–1604. <https://doi.org/10.1177/0956797621998312>
- Frey, R., & Fischer, O. (2025). Mapping the ecology of risk: 100 risky choices of modern life. *Psychological Science* 36(11), 846–861. <https://doi.org/10.1177/09567976251384975>
- Frey, R., Pedroni, A., Mata, R., Rieskamp, J., & Hertwig, R. (2017). Risk preference shares the psychometric structure of major psychological traits. *Science Advances*, 3(10), Article e1701381. <https://doi.org/10.1126/sciadv.1701381>
- Fruchterman, T. M. J., & Reingold, E. M. (1991). Graph drawing by force-directed placement. *Software: Practice and Experience*, 21(11), 1129–1164. <https://doi.org/10.1002/spe.4380211102>
- Galesic, M., Bruine de Bruin, W., Dalege, J., Feld, S. L., Kreuter, F., Olsson, H., Prelec, D., Stein, D. L., & van der Does, T. (2021). Human social sensing is an untapped resource for computational social science. *Nature*, 595(7866), 214–222. <https://doi.org/10.1038/s41586-021-03649-2>
- Gerrard, M., Gibbons, F. X., Houlihan, A. E., Stock, M. L., & Pomery, E. A. (2008). A dual-process approach to health risk decision making: The prototype willingness model. *Developmental Review*, 28(1), 29–61. <https://doi.org/10.1016/j.dr.2007.10.001>
- Gigerenzer, G. (2006). Out of the frying pan into the fire: Behavioral reactions to terrorist attacks. *Risk Analysis*, 26(2), 347–351. <https://doi.org/10.1111/j.1539-6924.2006.00753.x>
- Heatherston, T. F., Kozlowski, L. T., Frecker, R. C., & Fagerstrom, K.-O. (1991). The Fagerstrom Test for nicotine dependence: A revision of the Fagerstrom Tolerance Questionnaire. *British Journal of Addiction*, 86, 1119–1127. <https://doi.org/10.1111/j.1360-0443.1991.tb01879.x>
- Hogarth, R. M., Portell, M., & Cuxart, A. (2007). What risks do people perceive in everyday life? A perspective gained from the experience sampling method (ESM): A perspective gained from the experience sampling method. *Risk Analysis*, 27(6),

- 1427–1439. <https://doi.org/10.1111/j.1539-6924.2007.00978.x>
- Hogarth, R. M., Portell, M., Cuxart, A., & Kolev, G. I. (2011). Emotion and reason in everyday risk perception. *Journal of Behavioral Decision Making*, 24(2), 202–222. <https://doi.org/10.1002/bdm.689>
- Holt, C. A., & Laury, S. K. (2002). Risk aversion and incentive effects. *The American Economic Review*, 92(5), 1644–1655. <https://doi.org/10.1257/000282802762024700>
- Huang, Y., Luan, S., Wu, B., Li, Y., Wu, J., Chen, W., & Hertwig, R. (2024). Impulsivity is a stable, measurable, and predictive psychological trait. *Proceedings of the National Academy of Sciences*, 121(24), Article e2321758121. <https://doi.org/10.1073/pnas.2321758121>
- Hull, D. L. (1988). *Science as a process: An evolutionary account of the social and conceptual development of science*. University of Chicago Press. <https://doi.org/10.7208/chicago/9780226360492.001.0001>
- Hussain, Z., Binz, M., Mata, R., & Wulff, D. U. (2024). A tutorial on open-source large language models for behavioral science. *Behavior Research Methods*, 56(8), 8214–8237. <https://doi.org/10.3758/s13428-024-02455-8>
- Irwin, C. E., & Millstein, S. G. (1986). Biopsychosocial correlates of risk-taking behaviors during adolescence: Can the physician intervene? *Journal of Adolescent Health Care*, 7, S82–S96. [https://doi.org/10.1016/S0197-0070\(20\)30008-5](https://doi.org/10.1016/S0197-0070(20)30008-5)
- Jeffreys, H. (1961). *The theory of probability* (3rd ed.). Oxford University Press. <https://doi.org/10.1093/oso/9780198503682.001.0001>
- Johnson, E. J., Häubl, G., & Keinan, A. (2007). Aspects of endowment: A query theory of value construction. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 33(3), 461–474. <https://doi.org/10.1037/0278-7393.33.3.461>
- Josef, A. K., Richter, D., Samanez-Larkin, G. R., Wagner, G. G., Hertwig, R., & Mata, R. (2016). Stability and change in risk-taking propensity across the adult life span. *Journal of Personality and Social Psychology*, 111(3), 430–450. <https://doi.org/10.1037/pspp0000090>
- Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 263–292. <https://doi.org/10.2307/1914185>
- Klein, G. (2017). *Sources of power: How people make decisions* (2nd ed.). MIT Press. <https://doi.org/10.7551/mitpress/11307.001.0001>
- Knight, F. H. (1921). *Risk, uncertainty and profit*. Houghton Mifflin. <https://ssrn.com/abstract=1496192>
- Koch, U., & Cratsley, K. (2020). Psychological mechanisms. In V. Zeigler-Hill & T. K. Shackelford (Eds.), *Encyclopedia of personality and individual differences* (pp. 1–9). Springer International Publishing. https://doi.org/10.1007/978-3-319-28099-8_1562-1
- Kuhn, T. S. (1962). *The structure of scientific revolutions*. University of Chicago Press. <https://doi.org/10.7208/chicago/9780226458106.001.0001>
- Kuroda, K., Takahashi, M., & Kameda, T. (2023). Majority rule can help solve difficult tasks even when confident members opt out to serve individual interests. *Scientific Reports*, 13(1), Article 14836. <https://doi.org/10.1038/s41598-023-42080-7>
- Lee, M. D., & Wagenmakers, E.-J. (2014). *Bayesian cognitive modeling: A practical course*, Cambridge University Press. <https://doi.org/10.1017/CBO9781139087759>
- Lejuez, C. W., Read, J. P., Kahler, C. W., Richards, J. B., Ramsey, S. E., Stuart, G. L., Strong, D. R., & Brown, R. A. (2002). Evaluation of a behavioral measure of risk taking: The balloon analogue risk task (BART). *Journal of Experimental Psychology: Applied*, 8(2), 75–84. <https://doi.org/10.1037/1076-898X.8.2.75>
- Lesieur, H., & Blume, S. (1987). The south oaks gambling screen (SOGS): A new instrument for the identification of pathological gamblers. *The American Journal of Psychiatry*, 144, 1184–8. <https://doi.org/10.1176/ajp.144.9.1184>
- Lob, A. B., Fischer, O., & Frey, R. (2025, January 13). *A taxonomy of state-specific psychological mechanisms underlying decisions under risk and uncertainty*. <https://doi.org/10.31234/osf.io/a73y4>
- Loewenstein, G. F., Weber, E. U., Hsee, C. K., & Welch, N. (2001). Risk as feelings. *Psychological Bulletin*, 127(2), 267–286. <https://doi.org/10.1037/0033-2909.127.2.267>
- Loomes, G., & Sugden, R. (1982). Regret theory: An alternative theory of rational choice under uncertainty. *The Economic Journal*, 92(368), 805–824. <https://doi.org/10.2307/2232669>
- Makowski, D., Ben-Shachar, M., & Lüdtke, D. (2019). bayestestR: Describing effects and their uncertainty, existence and significance within the Bayesian framework. *Journal of Open Source Software*, 4(40), Article 1541. <https://doi.org/10.21105/joss.01541>
- Markiewicz, L., & Weber, E. U. (2013). DOSPERT's Gambling Risk-Taking Propensity Scale predicts excessive stock trading. *Journal of Behavioral Finance*, 14(1), 65–78. <https://doi.org/10.1080/15427560.2013.762000>
- Mellers, B., Schwartz, A., & Ritov, I. (1999). Emotion-based choice. *Journal of Experimental Psychology: General*, 128(3), 332–345. <https://doi.org/10.1037/0096-3445.128.3.332>
- Mishra, S., Barclay, P., & Sparks, A. (2017). The relative state model: Integrating need-based and ability-based pathways to risk-taking. *Personality and Social Psychology Review*, 21(2), 176–198. <https://doi.org/10.1177/1088868316644094>
- Murray, C. J. L., Abbafati, C., Abbas, K. M., Abbasi, M., Abbasi-Kangevari, M., Abd-Allah, F., Abdollahi,

- M., Abedi, P., Abedi, A., Abolhassani, H., Aboyans, V., Abreu, L. G., Abrigo, M. R. M., Abu-Gharbieh, E., Haimed, A. K. A., Abushouk, A. I., Acebedo, A., Ackerman, I. N., Adabi, M., ... Lim, S. S. (2020). Five insights from the global burden of disease study 2019. *The Lancet*, 396(10258), 1135–1159. [https://doi.org/10.1016/S0140-6736\(20\)31404-5](https://doi.org/10.1016/S0140-6736(20)31404-5)
- Nisbett, R. E., & Wilson, T. D. (1977). Telling more than we can know: Verbal reports on mental processes. *Psychological Review*, 84(3), 231–259. <https://doi.org/10.1037/0033-295x.84.3.231>
- Oberauer, K. (2022). The importance of random slopes in mixed models for Bayesian hypothesis testing. *Psychological Science*, 33(4), 648–665. <https://doi.org/10.1177/09567976211046884>
- Piper, M. E., Baker, T. B., Benowitz, N. L., Smith, S. S., & Jorenby, D. E. (2020). E-cigarette dependence measures in dual users: Reliability and relations with dependence criteria and E-cigarette cessation. *Nicotine & Tobacco Research*, 22(5), 756–763. <https://doi.org/10.1093/ntr/ntz040>
- Python Software Foundation. (2023). *Python language reference* (Version 3.10) [Computer software]. <https://www.python.org>
- R Core Team. (2024). *R: A language and environment for statistical computing*. <https://www.R-project.org/>
- Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In K. Inui, J. Jiang, V. Ng, & X. Wan (Eds.), *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)* (pp. 3982–3992). Association for Computational Linguistics. <https://doi.org/10.18653/v1/D19-1410>
- Samuelson, P. A. (1937). A note on measurement of utility. *The Review of Economic Studies*, 4(2), 155–161. <https://doi.org/10.2307/2967612>
- Saunders, J. B., Aasland, O. G., Babor, T. F., De La Fuente, J. R., & Grant, M. (1993). Development of the Alcohol Use Disorders Identification Test (AUDIT): WHO collaborative project on early detection of persons with harmful alcohol consumption-II. *Addiction*, 88(6), 791–804. <https://doi.org/10.1111/j.1360-0443.1993.tb02093.x>
- Savage, L. J. (1954). *The foundations of statistics*. Wiley.
- Schonberg, T., Fox, C. R., & Poldrack, R. A. (2011). Mind the gap: Bridging economic and naturalistic risk-taking with cognitive neuroscience. *Trends in Cognitive Sciences*, 15(1), 11–19. <https://doi.org/10.1016/j.tics.2010.10.002>
- Siegrist, M., & Cvetkovich, G. (2000). Perception of hazards: The role of social trust and knowledge. *Risk Analysis*, 20(5), 713–720. <https://doi.org/10.1111/0272-4332.205064>
- Simon, H. A. (1955). A behavioral model of rational choice. *The Quarterly Journal of Economics*, 69(1), 99–118. <https://doi.org/10.2307/1884852>
- Sirola, A., Kaakinen, M., Savolainen, I., & Oksanen, A. (2019). Loneliness and online gambling-community participation of young social media users. *Computers in Human Behavior*, 95, 136–145. <https://doi.org/10.1016/j.chb.2019.01.023>
- Skinner, H. A. (1982). The Drug Abuse Screening Test. *Addictive Behaviors*, 7(4), 363–371. [https://doi.org/10.1016/0306-4603\(82\)90005-3](https://doi.org/10.1016/0306-4603(82)90005-3)
- Slovic, P. (1987). Perception of risk. *Science*, 236(4799), 280–285. <https://doi.org/10.1126/science.3563507>
- Smithson, M., & Verkuilen, J. (2006). A better lemon squeezer? Maximum-likelihood regression with beta-distributed dependent variables. *Psychological Methods*, 11(1), 54–71. <https://doi.org/10.1037/1082-989X.11.1.54>
- Szrek, H., Chao, L.-W., Ramlagan, S., & Peltzer, K. (2012). Predicting (un)healthy behavior: A comparison of risk-taking propensity measures. *Judgment and Decision Making*, 7(6), 716–727. <https://doi.org/10.1017/S1930297500003260>
- Tetlock, P. E. (1992). The impact of accountability on judgment and choice: Toward a social contingency model. In M. P. Zanna (Ed.), *Advances in experimental social psychology* (pp. 331–376). Academic Press. [https://doi.org/10.1016/S0065-2601\(08\)60287-7](https://doi.org/10.1016/S0065-2601(08)60287-7)
- Vandekerckhove, J., Matzke, D., & Wagenmakers, E.-J. (2015). Model comparison and the principle of parsimony. In J. R. Busemeyer, Z. Wang, J. T. Townsend, & A. Eidels (Eds.), *The Oxford handbook of computational and mathematical psychology* (pp. 300–319). Oxford University Press. <https://doi.org/10.1093/oxfordhb/978019957996.013.14>
- Verschoor, A., D'Exelle, B., & Perez-Viana, B. (2016). Lab and life: Does risky choice behaviour observed in experiments reflect that in the real world? *Journal of Economic Behavior & Organization*, 128, 134–148. <https://doi.org/10.1016/j.jebo.2016.05.009>
- von Neumann, J., & Morgenstern, O. (1944). *Theory of games and economic behavior*. Princeton University Press.
- vor der Brück, T., & Pouly, M. (2019). Text similarity estimation based on word embeddings and matrix norms for targeted marketing. In J. Burstein, C. Doran, & T. Solorio (Eds.), *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: Human language technologies, volume 1 (long and short papers)* (pp. 1827–1836). Association for Computational Linguistics. <https://doi.org/10.18653/v1/N19-1181>
- Wagenmakers, E.-J., Lodewyckx, T., Kuriyal, H., & Grasman, R. (2010). Bayesian hypothesis testing for psychologists: A tutorial on the Savage–Dickey

- method. *Cognitive Psychology*, 60(3), 158–189. <https://doi.org/10.1016/j.cogpsych.2009.12.001>
- Weber, E. U., Blais, A.-R., & Betz, N. E. (2002). A Domain-Specific Risk-Attitude Scale: Measuring risk perceptions and risk behaviors. *Journal of Behavioral Decision Making*, 15(4), 263–290. <https://doi.org/10.1002/bdm.414>
- Wetzels, R., Matzke, D., Lee, M. D., Rouder, J. N., Iverson, G. J., & Wagenmakers, E.-J. (2011). Statistical evidence in experimental psychology: An empirical comparison using 855 *t* tests. *Perspectives on Psychological Science*, 6(3), 291–298. <https://doi.org/10.1177/1745691611406923>
- Wilke, A., Sherman, A., Curdt, B., Mondal, S., Fitzgerald, C., & Kruger, D. J. (2014). An evolutionary Domain-Specific Risk Scale. *Evolutionary Behavioral Sciences*, 8(3), 123–141. <https://doi.org/10.1037/ebbs0000011>
- Wolf, M., Kurvers, R. H. J. M., Ward, A. J. W., Krause, S., & Krause, J. (2013). Accurate decisions in an uncertain world: Collective cognition increases true positives while decreasing false positives. *Proceedings of the Royal Society B: Biological Sciences*, 280(1756), Article 20122777. <https://doi.org/10.1098/rspb.2012.2777>
- World Economic Forum. (2024). *The Global Risks Report 2024*. Retrieved February 1, 2024, from <https://www.weforum.org/publications/global-risks-report-2024/>

Received December 5, 2024

Revision received October 29, 2025

Accepted October 31, 2025 ■